		Target engagement	Dose	Efficacy	Safety
#15	**hepatitis C** HCV NS5A inhibitors (10)	**2011** HCV subgenomic replicon in Huh-7 cells, genotype 1a/1b and patient-derived NS5A hybrids	**2015** HCV subgenomic replicon in Huh-7 cells, genotype 1a/1b and patient-derived NS5A hybrids	**2020** HCV viral-kinetic model of on-treatment HCV RNA decline, later run in real time to set DAA duration	**2015** Rodent, dog and monkey tox Rat/monkey adrenal signal absent in patients
	hepatitis C NS5B non-nucleoside inhibitors (10)	**2014** HCV genotype 1 subgenomic replicon (Huh-7), with and without human serum	**2014** Rat and dog repeat-dose toxicology (dasabuvir)	**2018** Human-hepatocyte chimeric mouse, patient HCV, 4-week DAAs △ Replicon potent; filibuvir no SVR	**2014** Rat and dog repeat-dose toxicology (dasabuvir)
	hepatitis C HCV NS5B (nucleotide) inhibitors (7)	**2012** HCV subgenomic replicon (Huh-7 clone A; genotype 1b and chimeric 1-6 replicons)	**2013** Sofosbuvir rat/dog/mouse tox	**2013** HCV subgenomic replicon (Huh-7 clone A; genotype 1b and chimeric 1-6 replicons)	**2013** Mitochondrial toxicity assay HepG2/CEM assay cleared BMS-986094; heart failure
#16	**diabetic kidney disease** NR3C2 antagonists (3)	**2021** Rat urinary Na+/K+ natriuresis under MR-agonist challenge	**2021** Phase 2b population PK/exposure-response model of UACR, serum potassium and eGFR	**2020** Uninephrectomised DOCA-salt rat	**2021** Rat autoradiography and equinatriuretic DOCA-salt △ No rat hyperkalaemia; DKD had it
#17	**acute myeloid leukaemia** FLT3 inhibitors (4)	**2013** FLT3-ITD cell lines and MV4-11 xenografts (phospho-FLT3)	**2013** FLT3-ITD cell lines and MV4-11 xenografts (phospho-FLT3)	**2012** Ba/F3 FLT3-ITD resistance screen △ Combination: no survival gain	**2018** Rat and dog toxicology and safety pharmacology PRES and QT missed by tox species
	acute myeloid leukaemia CD33-directed depleters (3)	**2021** AML blasts + NK cells ex vivo NK activation unchanged despite CD33 saturation	**2017** CD33+ AML xenografts (mice)	**2025** Cord-blood CD33-CAR-NK: Molm13 xenografts, HSC colonies △ No single-agent responses in AML	**2025** Cord-blood CD33-CAR-NK: Molm13 xenografts, HSC colonies
#18	**malignant pancreatic neoplasm** MAP2K1/2 inhibitors (3)	**2012** Tumour xenografts, pERK by IHC (trametinib)	**2012** Tumour xenografts, pERK by IHC (trametinib)	**2018** RAS/BRAF lines, pimasertib KRAS status did not pick pimasertib responders	**2013** Rat, dog, rabbit, mouse safety Animals missed ocular harm, LVEF late
#19	**MASH** NR1H4 agonists (4)	**2019** Rodent FXR target-gene PD (tropifexor)	**2019** STAM and AMLN-diet NASH mice (tropifexor) No checkable prediction	**2021** STAM and AMLN-diet NASH mice (tropifexor) □ Outcome trial negative	**2024** High-fat-diet mouse ALT/cholesterol screen △ Cilofexor: no lipid change
	MASH FGF21 analogues (3)	**2020** Obese/diabetic cynomolgus monkeys	**2023** Pegozafermin population PK/PD exposure-response model	**2023** Pegozafermin population PK/PD exposure-response model △ Pegbelfermin failed on fibrosis	**2020** FGF21 variant carriers (UK Biobank) △ No mouse bone loss at 16 weeks
#20	**prostate cancer** AR antagonists (5)	**2013** LNCaP/AR castrate xenografts	**2013** LNCaP/AR castrate xenografts	**2016** LNCaP/AR castrate xenografts	**2019** Mouse brain:plasma distribution with rat and dog CNS toxicology (darolutamide)
	prostate cancer PARP1/2 inhibitors (5)	**2009** BRCA2-null V-C8, ES cells, xenografts; PARP inhibitor alone	**2018** AR-PARP cross-talk cells and xenografts	**2023** BRCA2-null V-C8, ES cells, xenografts; PARP inhibitor alone △ HRR-negative arm stopped, futile	**2023** Rat, dog repeat-dose tox of olaparib, niraparib, talazoparib △ Anaemia, neutropenia unforeseen
	prostate cancer CYP17A1/AR inhibitors (3)	**2018** VCaP mice, cyno/rat steroids	**2018** VCaP mice, cyno/rat steroids Exposure plateaued; PK stopped development	**2015** LAPC4 androgen-dependent xenografts in SCID mice, galeterone △ 1/17 PSA50 after enzalutamide	**2018** VCaP mice, cyno/rat steroids △ CNS tox closed prostate phase 2
	prostate cancer EZH2 inhibitors (3)	**2019** CRPC lines and xenografts, EZH2i + AR blockade	**2012** CRPC lines and xenografts, EZH2i + AR blockade No checkable prediction	**2026** CRPC lines and xenografts, EZH2i + AR blockade Patient result unclear	**2026** Rat and monkey repeat-dose toxicology (tazemetostat)
	prostate cancer FOLH1-directed depleters (3)	**2019** PSMA-high CRPC xenografts, PSMA ADC	**2011** PSMA-high CRPC xenografts, PSMA ADC No checkable prediction	**2022** PSMA-positive vs PSMA-negative prostate tumour-bearing mice	**2022** GLP single-dose rat, minipig and 4-week rat tox, cold PSMA-617 △ Low rodent PSMA hid xerostomia
#21	**follicular lymphoma** CD19-directed depleters (3)	**2011** Syngeneic anti-murine-CD19 CAR T cells vs 38c13 CD19+ lymphoma in C3H mice	**2015** Raji xenograft in NSG mice, CD19 CAR T No checkable prediction	**2024** Raji xenograft in NSG mice, CD19 CAR T	**2023** Humanized NSG mice, high B-ALL burden, CD19 CAR T cells △ CRS/neurotox despite lenzilumab
	follicular lymphoma PIK3CD inhibitors (3)	**2011** Patient B-cell tumour samples and B-cell lines ex vivo	**2024** Healthy-volunteer whole-blood basophil activation test, ex-vivo PI3Kδ PD assay for ME-401	**2014** Patient B-cell tumour samples and B-cell lines ex vivo	**2014** Rat and dog repeat-dose toxicology
#22	**ovarian cancer** PARP1/2 inhibitors (6)	**2009** Human tumour xenografts, needle-biopsy PAR ELISA (veliparib)	**2021** NOVA exposure–safety model (niraparib)	**2016** BRCA-deficient cancer cells and xenografts (niraparib)	**2017** Dog cardiovascular safety pharmacology (niraparib)
#23	**rheumatoid arthritis** BTK inhibitors (8)	**2013** BTK occupancy probe in CIA mice	**2022** CIA occupancy-efficacy PK/PD	**2019** Collagen-induced arthritis (rodent) CIA-active drug failed Phase 2	**2021** Human platelet aggregometry and humanized-mouse thrombus formation (acalabrutinib vs ibrutinib)
	rheumatoid arthritis SYK inhibitors (3)	**2014** Rat adjuvant- and collagen-induced arthritis (MK-8457)	**2018** Rat collagen-induced arthritis, therapeutic dosing from peak disease (lanraplenib/GS-9876)	**2014** Rat adjuvant- and collagen-induced arthritis (MK-8457) △ DAS28 no better than placebo	**2018** Cynomolgus monkey platelet/bleeding-time (lanraplenib) △ Serious infections missed, ended
	rheumatoid arthritis anti-CSF2 antibodies (3)	**2018** Mouse GM-CSF-IRF4-CCL17 pathway in zymosan/antigen arthritis pain models	**2025** Cynomolgus 13-week SC and 26-week IV GLP tox, otilimab △ Q2W regimen lost engagement	**2022** Mouse arthritis, anti-GM-CSF Failed in targeted-therapy-IR; met ACR20 elsewhere	**2025** Human anti-GM-CSF autoantibody alveolar proteinosis △ No PAP, 2915 patients, 2.5 years
#24	**multiple myeloma** TNFRSF17-directed depleters (8)	**2016** BCMA+ myeloma xenografts in NSG mice, CAR-T	**2026** Belantamab PopPK, tumour and keratopathy model No checkable prediction	**2019** BCMA+ myeloma xenografts in mice, ADC	**2020** Rat and rabbit repeat-dose toxicology (belantamab mafodotin)
	multiple myeloma PSMB5 inhibitors (3)	**2015** MM.1S myeloma xenografts in SCID mice, β5 PD, ixazomib	**2015** Ixazomib population PK, fixed 4 mg versus BSA-based dosing	**2015** MM.1S myeloma xenografts in SCID mice, β5 PD, ixazomib △ 9% PR vs marked PDX activity	**2015** Ixazomib GLP tox, 6-month rat and 9-month dog repeat dose △ Tolerated in dogs; GI tox ended it
#25	**melanoma** BRAF inhibitors (3)	**2010** BRAF V600E cell lines and xenografts	**2011** Colo205 BRAF V600E xenograft (exposure-response)	**2010** BRAF V600E vs wild-type cell-line panel (patient selection) △ INTERIM: intermittent worse again	**2014** HRAS Q61L keratinocytes and DMBA/TPA mouse skin △ FDA: cuSCC ~29%, animals blank
	melanoma anti-PDCD1 antibodies (3)	**2021** Humanised-PD-1 C57BL/6 mouse with S180 homografts, 124I-toripalimab immuno-PET	**2019** Human T-cell PD-1 occupancy in vitro (saturation threshold)	**2026** Human PD-1 knock-in mouse bearing MC38 tumours	**2024** Monkey 4/26-week GLP tox Endocrine and skin irAEs in melanoma cohort
#26	**Crohn disease** anti-IL23A antibodies (3)	**2018** T-cell transfer colitis mouse (anti-mouse IL-23p19 mAb)	**2022** Sponsor population-PK / exposure-response dose model from phase 2, risankizumab, mirikizumab	**2017** T-cell transfer colitis mouse (anti-mouse IL-23p19 mAb)	**2022** Monkey tox, risankizumab Hepatotoxicity in Crohn's; monkeys silent
#27	**open-angle glaucoma** ROCK inhibitors (3)	**2018** Normotensive monkey eye, topical netarsudil 0.04%	**2014** Normotensive monkey eye, topical netarsudil 0.04%	**2014** Normotensive monkey eye, topical netarsudil 0.04% △ Genotype did not predict response	**2017** Monkey 9-month topical ocular toxicology
Allergy	**allergic disease** allergen (immunotherapy) (5)	**2019** Patient PBMC Fel d 1-specific CD4+ T cells (ex vivo), Cat-PAD	**2019** Allergen exposure chamber	**2018** Mouse airway allergy, Fel d 1 Mouse protection, field loss to placebo	**2019** Allergen exposure chamber
	allergic rhinitis PTGDR2 antagonists (4)	**2013** Human whole-blood PGD2 eosinophil shape-change assay	**2014** Asthmatic allergen challenge	**2017** OVA mouse rhinitis Mouse nasal-symptom relief absent in Phase 3	**2017** Asthmatic allergen challenge
	allergic rhinitis allergen (immunotherapy) (8)	**2018** Conjunctival test, gp-ASIT+	**2019** Allergen exposure chamber	**2019** Conjunctival test, gp-ASIT+ Confirmatory field trial missed primary	**2019** Allergen exposure chamber
Autoimmune	**psoriasis** JAK1 inhibitors (3)	**2018** Human whole-blood cytokine-induced STAT phosphorylation, JAK1 selectivity	**2022** Rat, monkey repeat-dose tox	**2018** Rat adjuvant-induced arthritis (cross-indication; no psoriasis model was run)	**2022** Rat, monkey repeat-dose tox Dose-related platelet fall the animals missed
	psoriasis PDE4 inhibitors (3)	**2020** Human Th17 ex vivo skin model Not measured in patients	**2010** Mouse phorbol ear edema	**2019** In-vitro PDE4 enzyme and cellular assays, skin-delivery design △ Imiquimod mouse yes, patients no	**2011** Rabbit primary skin irritation and mouse local lymph node assay (2% ointment)
	psoriasis RORC inverse agonists (4)	**2021** Human whole-blood IL-17 secretion assay △ No IL-17 change in plaques	**2021** Human whole-blood IL-17 secretion assay	**2022** IL-23 minicircle mouse, oral BI 730357	**2015** Nonclinical toxicology species (not public) Model data unpublished
	psoriasis anti-IL17A antibodies (4)	**2016** Mice injected with human IL-17A, plasma KC (CXCL1) readout	**2016** Cynomolgus repeat-dose toxicology	**2023** Mice injected with human IL-17A, plasma KC (CXCL1) readout	**2016** People with inborn errors of IL-17 immunity